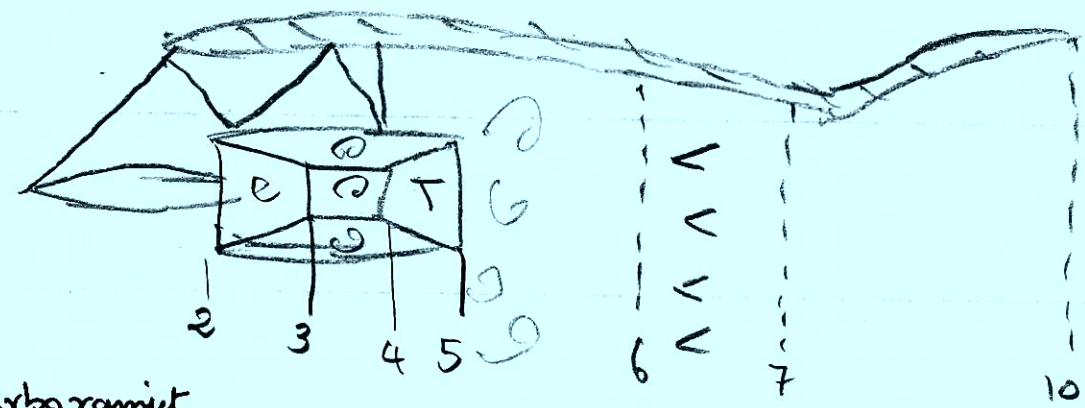
TRJ



Ideal Turboramjet

Simple 1D analysis ; $\eta_i = 1$; $c_p = \phi$

Core fluid : Define π_c as compression pr ratio

$$\pi_c = \frac{p_{t3}}{p_{t2}} = \frac{p_{t3}}{p_{t0}}$$

Since the compressor does work on the fluid, $p_{t3} > p_{t0}$

$$T_{t3} = T_{t0} \left(\pi_c\right)^{\frac{\gamma-1}{\gamma}}$$

Assume isobaric combustion in the core (good assumption esp because $M_3 \approx 0$)

$$p_4 = p_3$$

$\therefore M_3 \approx 0 \approx M_4 \qquad p_{t4} \approx p_{t3}$

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1st law of t/d in combustor

$$(1+f_{,c})T_{t4} = T_{t3} + \frac{f_{,c} H_{vf}}{c_p}$$

$$f_{,c} \equiv \frac{\dot{m}_{f,core}}{\dot{m}_{air,core}}$$

- $T_{t4} \equiv$ also turbine inlet temp.

$\dot{W}$ extracted by turbine $=$ $\dot{W}$ done on the fluid by compressor.

$$\bar{\dot{W}}_{comp} = -\bar{\dot{W}}_{turb} \quad \text{(no mechanical losses)}$$

1st law for turbine (adiabatic)

$$\bar{\dot{W}}_{turb} = \dot{m}\Delta h$$

$$= \dot{m}_t (h_{t5} - h_{t4}) \qquad \text{(assume } \Delta PE, \Delta KE, Q = 0\text{)}$$

1st law for comp.

$$\bar{\dot{W}}_{comp} = \dot{m}\Delta h$$

$$= \dot{m}_c (h_{t3} - h_{t0})$$

$$\dot{m}_t = \dot{m}_c + \dot{m}_{f,c}$$

To simplify, neglect $\dot{m}_{f,c}$ compared to $\dot{m}_c$.

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$$\because \quad \dot{W}_{comp} = -\dot{W}_{turb}$$

$$\cancel{\dot{m}_c}\,(h_{t3} - h_{t0}) = -\cancel{\dot{m}_c}\,(h_{t5} - h_{t4})$$

$$h_t = C_p T_t \qquad (C_p = ¢)$$

$$\boxed{T_{t5} = T_{t4} - (T_{t3} - T_{t0})}$$

T_{t5} : turbine exit temp

Ideal turbine $\Rightarrow$ no entropy production $\Rightarrow$ isentropic

$$\Rightarrow \boxed{P_{t5} = P_{t4}\left(\frac{T_{t5}}{T_{t4}}\right)^{\gamma/\gamma-1}}$$

And w/ T_{t5} & P_{t5}, can calculate ΔKE & thrust for turbo operation

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For ramjet fluid :

- Ideal $\Rightarrow$ No losses $\Rightarrow \eta_i s = 1$
- No work done, & so to increase in P_t
- Define station 6, as the end of the mixing region (ram & core fluid are well-mixed) & let station 6 be located w/s of fuel injectors (see figures earlier)

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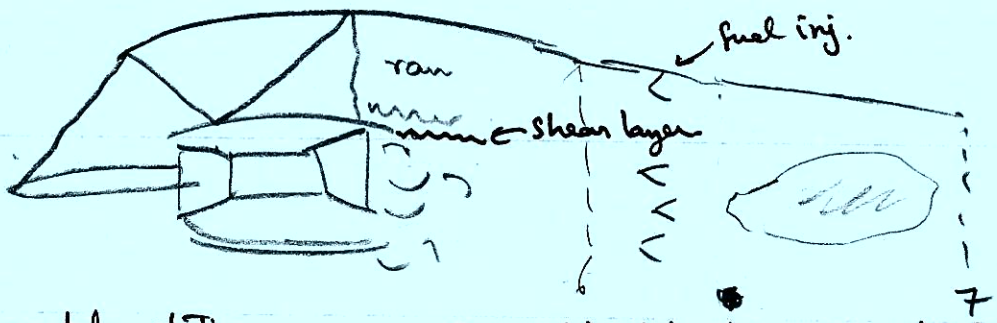

- Need to match static pressures & would like to match velocities, but the latter is not possible.

$$T_{t6} \cong \frac{\dot{m}_c T_{ts} + \dot{m}_r T_{to}}{\dot{m}_a}$$

total incoming mass flux.

$\dot{m}_r \equiv$ ram fluid

$\dot{m}_c \equiv$ core "

$$p_{t6} \cong \frac{p_{ts} + p_{to}}{2}$$

- Across fuel injectors : isobaric & low Mach #

$$p_{t07} \cong p_{t6}$$

- 1st law for fuel inj & burner.

$$(\dot{m}_c + \dot{m}_r + \dot{m}_{f,r}) T_{t7} = (\dot{m}_c + \dot{m}_r) T_{t6} + \frac{\dot{m}_{f,r} H_{vf}}{c_p}$$

$$f_{,r} = \frac{\dot{m}_{f,r}}{\dot{m}_r}$$

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So,

$$\left(1 + \frac{f_{,r}}{(BPR)^{-1}+1}\right) T_{t7} = T_{t6} + \frac{f_{,r}}{\left(BPR^{-1} + 1\right)} \frac{H_{vt}}{C_p}$$

$$0 < BPR < \infty$$

1st Law for nozzle.

$$h_{t,10} = h_{10} + \frac{1}{2} V_{10}^2 \qquad (10 \equiv \text{exit})$$

$$h_{t10} = C_p T_{t7}$$

$$h_{10} = C_p T_{10}$$

$$\frac{T_{10}}{T_{t10}} = \frac{T_{10}}{T_{t7}} = \left(\frac{p_{10}}{p_{t10}}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{p_{10}}{p_{t6}}\right)^{\frac{\gamma-1}{\gamma}}$$

$$V_{10} = \left\{ 2 C_p T_{t7} \left(1 - \left(\frac{p_{10}}{p_{t6}}\right)^{\frac{\gamma-1}{\gamma}}\right) \right\}^{1/2}$$

↳ Check / derive

$$f = \frac{\dot{m}_f}{\dot{m}_a} = \frac{\dot{m}_{f,c} + \dot{m}_{f,r}}{\dot{m}_c + \dot{m}_r} = \frac{f_c}{1 + BPR} + \frac{f_r}{BPR^{-1} + 1}$$

$$\boxed{\frac{\tau}{\dot{m}_a} = (1+f) V_{10} - V_0}$$

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Eg: For $\Pi_c = 20$, @ what min Mach# shd BPR $\to \infty$?

BPR $\to \infty \Rightarrow$ pure ramjet.

Ideal ramjet $\Rightarrow$ no losses during compression $\Rightarrow p_t$ remains the same.

At the burner entry $M_3 \approx 0 \Rightarrow p_3 = p_{t3} = p_{t2} = p_2$

$$\Pi_c = \frac{p_{t2}}{p_0}$$

$\because$ no losses from compression $p_{t2} = p_{t0}$

$$\Rightarrow \frac{p_{t0}}{p_0} = \Pi_c = 20$$

$$\left(1 + \frac{\gamma-1}{2} M_0^2\right)^{\gamma/\gamma-1} = 20$$

$$\boxed{M_0 = 2.6}$$

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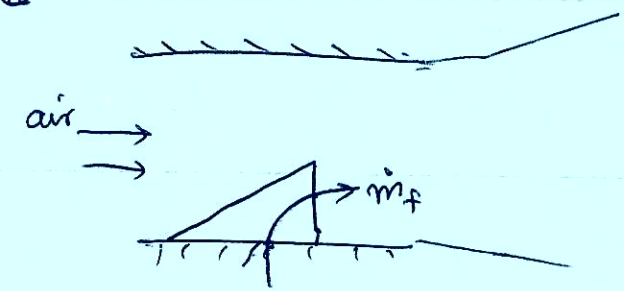
Scramjet - Rocket : As seen in plots of τ vs M, @ high M, $\tau \to 0$, $I_{sp} \to 0$.

To achieve orbit, need high speed ~ $M_0 = 25$.

To maintain reasonable η_0 @ high altitude is very hard because $[O_2]$ down to 10^{-4} in upper atm. ⇒ very hard to sustain combustion.

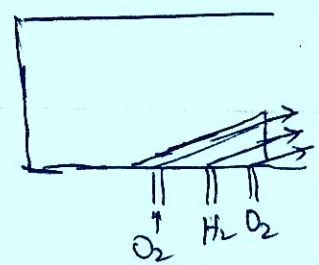
⇒ Switch to rockets.

Scramjet - rocket combine mode



To switch to rocket mode, close off the inlet & inject oxidizer w/ fuel.

→ Complexity is mainly w/ going from turbo to ramjet & ramjet to scramjet.



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